

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

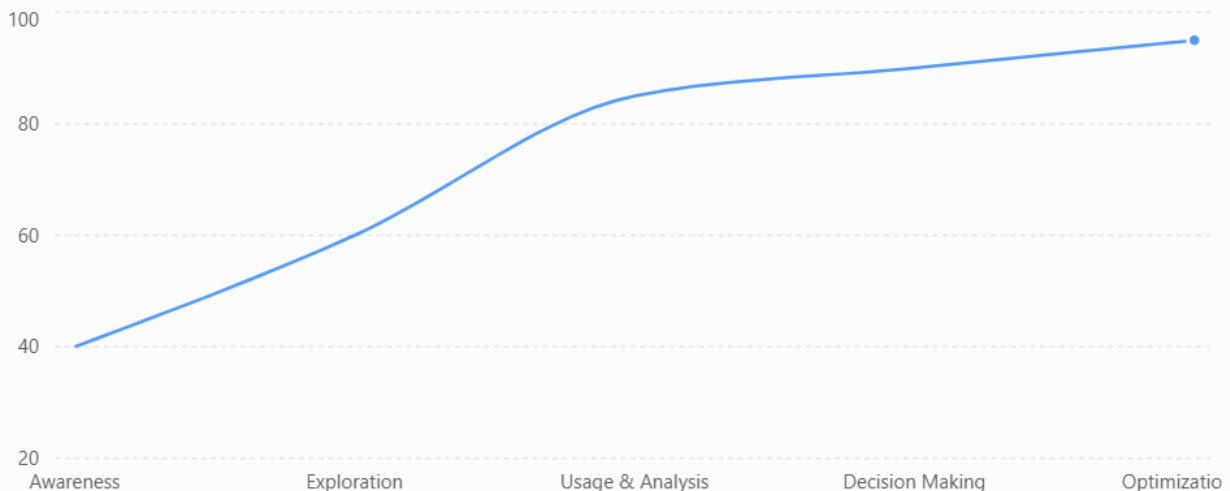
TEAM ID	SWTID-2026-6767
PROJECT TITLE	Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	2 MARKS

CHAPTER – III REQUIREMENT ANALYSIS

3.1 – Customer Journey Map

Customer engagement across journey stages

Illustrative engagement level as users progress through the EV charge and range analysis tool journey.



The Electric Vehicle user experience is shown in the Customer Journey Map when they use the Tableau-based visualization tool. The main goal of this tool is to help Electric Vehicle users keep track of their battery charge levels look at their driving range find charging patterns and make decisions about charging and planning trips.

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The journey starts with the Awareness stage, where Electric Vehicle users realize they need a way to better understand how their Electric Vehicles battery works and how it charges. They may have problems like worrying about how they can drive, not knowing if charging stations are available or having trouble understanding the information from their vehicle. They find out about the Tableau-based visualization tool from websites, media and Electric Vehicle groups.

In the Exploration stage. Electric Vehicle users look at what the tool can do. They try out the dashboards look at example visualizations and learn how the tool can turn raw Electric Vehicle data into useful information. The tool has to-understand charts, maps and battery information that makes Electric Vehicle users more interested and confident in using the tool.

When Electric Vehicle users are in the Usage and Analysis stage they add their Electric Vehicle data to the Tableau dashboards. The tool shows them how much charge is left in their battery how they charge how much energy they use and how far they can still drive. The tool also has filters and features that let Electric Vehicle users look deeper at how their vehicle performs in driving conditions, which helps them understand how well their vehicle uses energy.

The Decision-Making stage is about using the information to make decisions. Electric Vehicle users use the tool to plan trips make charging schedules and reduce the risk of running out of battery. The visualization tool helps Electric Vehicle users make decisions based on data, which makes driving more convenient and gives them confidence.

Finally, in the Optimization and Advocacy Stage Electric Vehicle users keep improving how they drive and charge their vehicles based on what they learn from the dashboards. As they get better, at managing their battery and do not worry much about running out of charge they tell other Electric Vehicle owners, groups and people about the solution.